

Cross-Model Hidden State Analysis

Caving Under Pressure — CS6120 NLP Project | Qwen2.5-7B · Llama-3.1-8B · Gemma-2-9B | April 2026

1. MODEL SUMMARY

Metric	Qwen2.5-7B	Llama-3.1-8B	Gemma-2-9B
Questions	153	146	176
Flipped	149 (97.4%)	49 (33.6%)	174 (98.9%)
Held firm	4	97	2
Mean flip turn	3.77	~3.0	1.77
L2 trend under pressure	Rises (+11.5 peak)	Rises (+0.83 peak)	Drops (−7.4%)
CosSim T0→T1	0.835	0.824	0.793
Dominant layers	Layer 28 (last)	Layers 30–32 (last 3)	Late layers
Architecture	29 layers, 3584 dims	33 layers, 4096 dims	42 layers, 3584 dims

2. CROSS-MODEL CONCLUSIONS

Conclusion 1 — CosSim T0→T1 is the most robust cross-model signal

Across all three models, the largest hidden state directional shift occurs at the very first pressure prompt — not at the flip turn, and not at Turn 4 where most behavioral changes occur. The model internally registers the conflict immediately.

Model	CosSim T0→T1	Mean Flip Turn
Gemma-2-9B	0.793	1.77 (earliest)
Llama-3.1-8B	0.824	~3.0
Qwen2.5-7B	0.835	3.77 (latest)

Lower T0→T1 cosine similarity = more internal disruption at first pressure = earlier behavioral flip. This directional relationship holds across all three model families and is the strongest cross-model claim in this study.

Conclusion 2 — L2 direction is model-dependent; late-layer concentration is universal

Qwen and Llama show **rising** L2 norms under pressure (activation amplification — processing the conflict). Gemma shows **dropping** L2 norms (activation suppression — representation collapse). L2 alone is therefore not a universal sycophancy indicator.

However, in all three models, the sycophancy-relevant signal is localized to the **last 2–4 transformer layers**: Layer 28 in Qwen (accounts for ~95% of the signal), Layers 30–32 in Llama (top drifters T0→T5),

and late layers in Gemma. Sycophancy is a task-level output decision — architecturally, these live in the final layers.

The late-layer concentration finding is consistent across model families, supporting the interpretation that sycophancy has a common architectural locus even when scale-dependent activation dynamics differ.

Conclusion 3 — Llama-8B is the only model with a usable control group

Qwen (97.4% flip) and Gemma (98.9% flip) provide almost no non-flipped questions for comparison. Llama's 33.6% flip rate gives 49 flipped vs 97 non-flipped — a near 1:2 ratio that supports a proper classification task.

Model	Best suited for
Qwen-7B, Gemma-9B	Pre-flip trajectory analysis (watching hidden states build before a flip)
Llama-8B	Flipped vs non-flipped classification (do Turn 0/1 hidden states predict whether a question will ever flip?)

3. PROBE TRAINING STRATEGY

Each model requires its own probe — hidden dimensions and layer counts are incompatible across models. Train and evaluate independently, then compare accuracy as the cross-model generalizability claim.

Probe A — Qwen2.5-7B · Layer 28 · 3584 dims

TASK

Given hidden state at Turn N (pre-flip window), predict whether a flip occurs at Turn N+1.

TRAINING DATA

149 flipped questions × 5 turns = 745 samples. Label = 1 if flip at next turn, 0 otherwise.

VALIDATION

Leave-one-question-out cross-validation.

PROVES

Pre-flip hidden states in the primary model carry detectable pre-behavioral signal.

Probe B — Llama-3.1-8B · Layers 30–32 · 4096 dims

TASK

Given hidden state at Turn 0 or Turn 1, predict whether this question flips at any point across 5 turns.

TRAINING DATA

49 flipped + 97 non-flipped = 146 questions. One hidden state vector per question.

VALIDATION

Stratified 5-fold cross-validation (preserves flip/no-flip ratio). Train separate probes on Layers 30, 31, 32 and compare accuracy.

PROVES

The internal state at first pressure already encodes information about eventual behavioral outcome — the early-warning claim, stronger than pre-flip detection alone.

Probe C — Gemma-2-9B · Late layers · 3584 dims

TASK

Same as Probe A, but note: mean flip at T1.77 means most pre-flip data is just Turn 0. Frame explicitly as "Turn 0 hidden state → predicts flip at Turn 1 or 2."

TRAINING DATA

174 flipped questions. Identify top 3 drifting layers from the layer drift plot and train on those.

CAUTION

Gemma's early flipping creates class imbalance in the pre-flip window. Report this as a limitation and apply class weighting if needed.

Target accuracy and cross-model claim

Probe	Model	Layer(s)	Target Accuracy
A	Qwen-7B	28	> 70%
B	Llama-8B	30–32	> 70%
C	Gemma-9B	TBD (top drift layers)	> 70%

If all three probes exceed 70% accuracy independently, the claim is: linear probes trained on late-layer hidden states can predict sycophantic behavior across model families, suggesting the pre-flip internal signal is a general property of transformer-based LLMs under social pressure.

4. COSSIM T0→T1 AS A MODEL-AGNOSTIC FEATURE

Because the T0→T1 cosine similarity drop is the most consistent signal across all three models, it can be included as a derived feature alongside raw hidden states — enabling a clean ablation of whether engineered metrics improve probe accuracy over raw vectors alone.

```
# Feature engineering for probes
cos_t0_t1 = cosine_similarity(hidden_state[turn_0], hidden_state[turn_1])
l2_delta  = l2_norm(hidden_state[turn_1]) - l2_norm(hidden_state[turn_0])

# Stack as supplementary features alongside the hidden state vector
probe_features = np.concatenate([hidden_state_vector, [cos_t0_t1, l2_delta]])
```